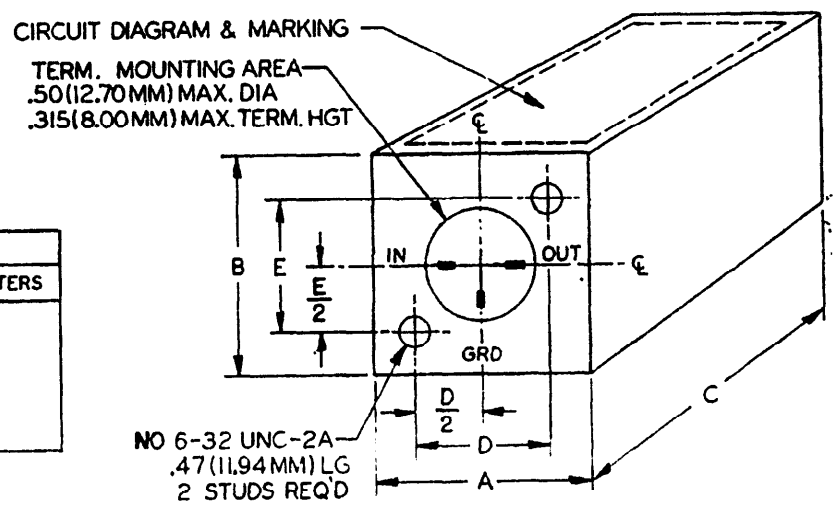
MILITARY SPECIFICATION SHEET

DELAY LINE, PULSE, ELECTROMAGNETIC, FIXED

STYLE PB

The complete requirements for procuring the delay line described herein shall consist of this document and the latest issue of Specification MIL-D-23859.

DIMENSIONS		
LTR	INCHES	MILLIMETERS
A	1.19	29.73
B	1.19	29.73
C	2.00	50.80
D	.688	17.48
E	.688	17.48



NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are .xx ± .03 (.76 mm) and .xxx ± .015 (.38 mm).

FIGURE 1. Case and mounting dimensions.

Type designation, terminating impedance, delay time, attenuation and rise time.

Part number	Type designation	Nominal terminating impedance (ohms)	Delay (usec)	Attenuation (db max)	Max rise time (usec)
M23859/02-001	DL500PB2506C	50	0.25	1.0	.045
M23859/02-002	DL500PB5006C	50	0.50	1.0	.089
M23859/02-003	DL500PB1005C	50	1.0	1.0	.178
M23859/02-004	DL101PB1005C	100	1.0	1.0	.178
M23859/02-005	DL201PB2505C	200	2.5	3.0	.446
M23859/02-006	DL501PB5005C	500	5.0	3.0	.890
M23859/02-007	DL202PB5005C	2,000	5.0	3.0	.890
M23859/02-008	DL501PB1004C	500	10.0	6.0	1.79
M23859/02-009	DL102PB1004C	1,000	10.0	6.0	1.79
M23859/02-010	DL202PB1004C	2,000	10.0	6.0	1.79

FSC 5840

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ENGINEERING DATA:

Case material: Ferrous type, hot tinned or fuzed electrotinned.

Finish: Grey enamel in accordance with MIL-E-15090, type I, class 2.

Weight: 7 ounces maximum.

Terminals: Three hooked type solder lugs capable of accepting two No. 22 AWG buss leads minimum.

Input pulse:
Peak pulse voltage: 150 volts peak maximum.
Pulse width: Shall be greater than three times the delay line pulse rise time and less than twice the delay.
Pulse rise time: 15 percent to 20 percent of delay line rise time.

Nominal terminating impedance: See table.

Style: PB.

Figure of merit: C.

REQUIREMENTS:

Design and construction: See figure 1.

Enclosure: See figure 1.

Delay time: See table.

Rise time: See table.

Attenuation: See table.

Distortion: ±10 percent at maximum delay.

Dielectric withstanding voltage: Method 301 of MIL-STD-202.

Delay time variation with temperature: ±0.5 percent from -55°C to +95°C.

Insulation resistance: Method 302, test condition A, MIL-STD-202. 1,000 megohms minimum.

Thermal age stability: 3 percent maximum delay change permissible. Rise time, attenuation and distortion shall still meet the requirements of this specification sheet.

Sealing: As specified in 4.7.3.1.

Terminal strength: Method 211 of MIL-STD-202.
Test condition A - Applied force 4-1/2 pounds.
Test condition C - Applied force 5 pounds.

Salt atmosphere: Method 101, test condition B, MIL-STD-202.
Salt solution - 5 percent.
Measurements - Not applicable.

Pressure: Method 105, test condition C, MIL-STD-202. Perform dielectric withstanding voltage test during reduced pressure and at 250 vdc.
Method of mounting - Normal.

Vibration: Method 204, test condition B, MIL-STD-202.
Measurements - Not applicable.

Shock: Method 205, test condition A, MIL-STD-202.

Thermal shock: Method 107, test condition A, MIL-STD-202, except that upper temperature limit is 95°C.
Measurements - Dielectric withstanding voltage and insulation resistance.

Immersion: Method 104, test condition A, MIL-STD-202.
Measurements - Insulation resistance.

Moisture resistance: Method 106 of MIL-STD-202.
Loading voltage - Not applicable.
Polarizing voltage - 100 volts dc.
Measurements - Insulation resistance and dielectric withstanding voltage.

Flammability: Not applicable.

Life: Method 108, test condition D, MIL-STD-202.
Test temperature - 95° ±2°C.
Test voltage - 150 vdc applied.
Measurements - Dielectric withstanding voltage, insulation resistance, delay time variation with temperature and thermal age stability.

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Part number: Consists of the basic number of this specification sheet and a dash number taken from table.

Example: M23859/02-001

Basic document number

Dash number from table

Custodians:
Army - EL
Navy - SH
Air Force - 80

Review activities:
Army - MU, EL
Navy - SH, AS, OS
Air Force - 80, 17, 85

User activities:
Army - MI
Navy - MC
Air Force - NONE

Preparing activity:
Navy - SH

(Project 5840-0029)

Code "C"

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